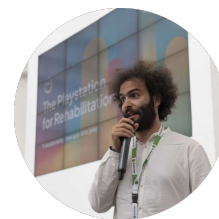


Giovanni Tauro

Design Engineer, MSc | Product Development & Interactive Systems

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🌐 LinkedIn 🐙 GitHub 🎓 Scholar 📁 Portfolio 🛠️ GrabCAD



R&D Product Engineer and PhD Candidate with 5+ years of experience in applied research, functional prototyping, IoT systems, and CE product development. Experienced in cross-functional projects spanning embedded electronics, software development, wearable technologies, indoor localization, and multimodal feedback systems. Driven by taste, I combine human-centered design, sensor-based interaction, and engineering to deliver innovative solutions for healthcare technologies.

Work Experience



R&D Product Engineer, STIIMA, Lecco

Feb 2021 – Present

- Developed sensor-based rehabilitation and assistive technology systems, translating clinical requirements and user needs into functional prototypes for experimental validation .
- Defined the architecture of multimodal IoT feedback platforms, integrating wearable sensing, indoor localization, BLE communication, and adaptive audio, visual, and haptic feedback.
- Delivered 3+ functional prototypes across embedded electronics, wearable devices, and interactive systems, supporting projects from concept development to ISO compliance technology delivery.
- Designed communication infrastructures and evaluation protocols for digital health applications, assessing usability, system performance, and user interaction through experimental testing.
- **Head of Product (Co-Founder), Rehabilia Technologies s.r.l.** (May 2023 – Present), applying this R&D expertise to lead product development as Head of Product:
 - Led end-to-end product development of rehabilitation technologies, from user research and problem discovery to validation and product delivery.
 - Defined product requirements and priorities by translating user needs, business goals, and stakeholder feedback into actionable roadmaps.
 - Collaborated with engineering, design, and clinical stakeholders to align product strategy and deliver iterative product improvements.
 - Validated product decisions through usability testing, interviews, and data-driven loops.

Embedded Systems • IoT • ESP32 • Arduino • PlatformIO • VS • BLE GATT • ROS2 • C++
• C# • Python • Prototyping • 3DCAD • FDM • Material Selection • UX Research • Product Development • Stakeholder Communication • ASANA • Cross-functional Teamwork • Project Management • Public Speaker



Freelancer, FLORENCE

Jan 2016 – Sep 2017, collaboration with **DIMA3D**

- Delivered end-to-end reverse engineering workflows, converting 3D scan data into validated parametric CAD models and production-ready assets.

CAD • Reverse Engineering • Geomagic • SolidWorks • 3D Scanning • Product Development
• Technical Documentation • Manufacturing Preparation

Selected Projects



[GIT_1](#)
[GIT_2](#)
[GIT_3](#)

OmniTrack: IoT-based Multimodal Navigation System /Frontiers

- Built end-to-end hardware/software prototypes, validated with 16 blind participants, integrating IoT-oriented sensing, processing, and user feedback into a complete system.
- Applied real-time data processing and system integration principles to transform physical-world signals into actionable information.

Python • Data Analysis • C++ • JS • Dashboard • ESP32 • ROS2 • BLE GATT • Sensors • I2C • Linux



PhiCube: reconfigurable bilateral robotic device for neurorehabilitation /Frontiers

- Designed a modular rehabilitation device combining UX research, mechanical design, embedded systems, and multimodal interaction.
 - Validated with 19 children (SUS 85.8/100); 134 patients in two on-going multicentric studies.
- UX • Interaction Design • CAD • Prototyping • Assembly • Usability

RehaBalloon: Interactive Platform



[GIT](#)

- Prototyped a low-cost rehabilitation device in 24 hours during the LMU Munich "Hacking with Hardware" challenge, integrating pressure sensing, visual feedback, and embedded electronics.
- Implemented a TensorFlow Lite model on Arduino embedded machine learning classifier exercise patterns and record rehabilitation activity metrics in real time.

Seeed Studio • Sensors • C++ • TensorFlow • Machine Learning

Education

2023–27 **PhD, Mechanics and Advanced Engineering Sciences** *Bologna University, IT*

Topic: IoT-based Multimodal Navigation System – OmniTrack

- Research Mobility, [Utrecht University](#), NLD: Sensory navigation and experimental psychology.
- Research Mobility, [Toyo University](#), JP: Feedback algorithm and behaviour analysis.

2017–20 **MSc, Design&Engineer** *Politecnico di Milano, IT*

Topic: UX development of an ultrasonic dental device: from concept to validation: WashB

- Research Mobility, [Chiba University](#), JP: UX research studies for Fujitsu and Panasonic.

2012–15 **BSc, Industrial Design** *Florence University, IT*

Topics: Foundation in research methodologies, development, and critical thinking.

Certifications

2026 **Claude 101 Badge — Claude Code Badge — Claude** *Anthropic, Online*

2024 **.ppt — Robotics and Allied Technologies for Rehab** *MUR, Rome, Italy*

Patent

2023 Malosio, M., Nicora, M. L., Redaelli, D., Tauro, G. [WO2023214349A1](#) *Device for bilateral rehabilitation.*

Languages

Italian: Native

English: Advanced (IELTS 2017)

French: Basic (A2)

Lecco, October 9, 2026

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